



Yeshvanth Reddy Guntaka

ML / AI · ROBOTICS & AUTONOMY · REAL-TIME SYSTEMS

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SUMMARY

Engineer across ML, computer vision and robotics. Architected a 12-layer autonomous drone software stack as CTO, and independently built eight open-source ML systems spanning pose estimation, 3D reconstruction, multi-agent orchestration and real-time streaming. I keep the hard part provable - pure, unit-tested cores with hardware and models injected at the edges - and ship the whole product: API, web and mobile clients, CI and containers.

EXPERIENCE

Data Analyst VJ Creative Solutions LLC

Jan 2026 - Present · Casper, WY

- Translate business questions into data-driven analyses, delivering insights via SQL queries, dashboards and visualizations.
- Design and maintain ETL/ELT pipelines with data-quality checks, improving integration accuracy and reducing downstream errors.
- Automate infrastructure deployment using Ansible/SaltStack and containerized services, cutting manual configuration time.

Chief Technology Officer Entriya LLC

Jul 2025 - Dec 2025 · Boston, MA

- Architected a 12-layer autonomous drone software stack with standards for modularity, observability and scalability.
- Built high-performance ROS2 modules for sensor fusion, world modeling and obstacle avoidance.
- Integrated MPC, EKF and consensus algorithms into an intent-to-command autonomy pipeline.
- Directed technical strategy and cross-functional engineering execution, improving system reliability.

Software Developer Intern Entriya LLC

Feb 2025 - Jul 2025 · Boston, MA

- Engineered EKF-based state estimation and obstacle-avoidance logic, improving robustness under sensor noise and latency.
- Implemented sensor-fusion and real-time messaging components, enabling lower-latency perception and navigation.
- Addressed embedded constraints through micro-kernel command arbitration and performance-focused design.

Data Science Intern Inmovidu Tech

May 2020 - Jul 2020 · Hyderabad, India

- Built and evaluated ML models using scikit-learn and TensorFlow with feature engineering and hyperparameter tuning.
- Performed end-to-end data preparation and exploratory analysis with Python, Pandas and NumPy.
- Developed a real-time speech-to-text / text-to-speech assistant using NLP libraries in a production pipeline.

SELECTED OPEN-SOURCE PROJECTS

AI Digital Tailor MediaPipe · FastAPI · NumPy · Flutter · Docker

83.8% holdout accuracy

Body measurements from phone photos: height-calibrated landmarks with an elliptical torso model for chest/waist/hip, a logistic-regression size classifier with calibrated probabilities, parametric 3D avatar export, real-time guided web capture and a Flutter mobile client.

AI Workout Coach MediaPipe · OpenCV · SQLite · Streamlit

8 exercises, 29 tests

Hysteresis rep-counting state machines over joint-angle geometry with per-rep depth scoring and live form cues; workout mode switches exercises mid-session and a dashboard tracks daily volume.

Gesture-Based Coding Assistant MediaPipe Hands · Speech · Codegen

6 gestures, 9 intents

Rotation-invariant finger-state detection with edge-triggered gesture smoothing, plus a voice-command grammar that compiles spoken instructions into Python or JavaScript injected into the focused IDE.

Multi-Agent Orchestrator & Stream Engine Redis Pub/Sub · Redis Streams · FastAPI

5 agents, tiered alerting

Five coordinating agents that plan, execute, monitor and recover tasks autonomously; plus a streaming engine with consumer groups and threshold + z-score anomaly detection.

EDUCATION

M.S., Computer Information Systems

New England College, Henniker, NH

Dec 2024 · GPA 3.67

B.Tech, Computer Science

Anil Neerukonda Institute of Technology and Sciences, Visakhapatnam, India

Apr 2021 · GPA 3.85

TECHNICAL SKILLS

Languages Python, Java, C, C++, R, JavaScript, Dart, SQL

ML & Vision TensorFlow, scikit-learn, MediaPipe, OpenCV, Pandas, NumPy, time-series forecasting, 3D geometry

Robotics ROS2, sensor fusion, EKF, MPC, obstacle avoidance, agent-based modeling, swarm intelligence

Backend FastAPI, Django, Flask, Node.js, REST API design, Redis (Streams/PubSub), SQLAlchemy, MySQL, SQLite, Firebase

Frontend/Mobile React, Vite, Tailwind CSS, Flutter (cross-platform), Streamlit, Android

DevOps & Tools Docker, GitHub Actions, Ansible/SaltStack, Git, pytest, Linux